Experiment 14: Diet Recommendation Expert System

Aim:

To develop a rule-based expert system that recommends personalized diet plans based on age, BMI and health condition.

Algorithm:

1. Take the age, BMI and health condition of the person as input.
2. Apply the condition rules (diabetes, hypertension, heart disease) to select a condition-specific diet.
3. Apply the BMI rules (underweight, normal, overweight, obese) to select a weight-related diet.
4. Apply the age rules (teenager, senior) to add age-specific advice.
5. Combine all fired rules and display the recommended diet plan.

Code:

```
def recommend_diet(age, bmi, condition):
    rules = []

    if condition == "diabetes":
        rules.append("Low sugar, high fibre diet: whole grains, vegetables, lentils")
    elif condition == "hypertension":
        rules.append("Low sodium DASH diet: fruits, vegetables, low-fat dairy")
    elif condition == "heart disease":
        rules.append("Low saturated fat diet: oats, nuts, fish, olive oil")

    if bmi < 18.5:
        rules.append("Underweight: high-calorie, protein-rich diet (milk, eggs, nuts, paneer)")
    elif bmi < 25:
        rules.append("Normal weight: balanced diet with carbohydrates, proteins and healthy fats")
    elif bmi < 30:
        rules.append("Overweight: calorie-controlled diet with more vegetables and regular exercise")
    else:
        rules.append("Obese: strict low-calorie diet, avoid fried and sugary food, consult a dietician")

    if age < 18:
        rules.append("Teenager: include calcium and iron rich foods for growth")
    elif age > 60:
        rules.append("Senior: soft, easily digestible food with extra calcium and vitamin D")

    return rules

patients = [
    ("Ravi", 45, 31.2, "diabetes"),
    ("Anita", 16, 17.5, "none"),
    ("Mohan", 65, 26.4, "hypertension"),
]

for name, age, bmi, cond in patients:
    print(f"Patient: {name} | Age: {age} | BMI: {bmi} | Condition: {cond}")
    for r in recommend_diet(age, bmi, cond):
        print("  -", r)
    print()
```

Sample Output:

Output

```
Patient: Ravi | Age: 45 | BMI: 31.2 | Condition: diabetes
  - Low sugar, high fibre diet: whole grains, vegetables, lentils
  - Obese: strict low-calorie diet, avoid fried and sugary food, consult a dietician

Patient: Anita | Age: 16 | BMI: 17.5 | Condition: none
  - Underweight: high-calorie, protein-rich diet (milk, eggs, nuts, paneer)
  - Teenager: include calcium and iron rich foods for growth

Patient: Mohan | Age: 65 | BMI: 26.4 | Condition: hypertension
  - Low sodium DASH diet: fruits, vegetables, low-fat dairy
  - Overweight: calorie-controlled diet with more vegetables and regular exercise
  - Senior: soft, easily digestible food with extra calcium and vitamin D
```

Result:

The expert system fired the relevant IF-THEN rules for each person and produced a personalized diet plan combining condition, BMI and age advice.